



United International University (UIU)
Department of Computer Science and Engineering
CSE 1325: DIGITAL LOGIC DESIGN, Final Summer 2025

Total Marks: **40** Duration: 2 hours

Answer ALL Questions

**Any examinee found adopting unfair means will be expelled from the
trimester/program as per UIU disciplinary rules.**

1. You are designing an energy management controller for a smart greenhouse. The system has two internal control states represented by T flip-flops T1 and T2, and one external control input X that indicates whether the system is in active monitoring mode ($X = 1$) or idle mode ($X = 0$). The state of the controller determines how different environmental devices (like fans, heaters, and lights) operate.

The system behaves as follows:

- When $X = 0$, the system maintains its current state (no transitions).
- When $X = 1$, the controller cycles through the following sequence of states to adjust environmental conditions dynamically: $00 \rightarrow 11 \rightarrow 10 \rightarrow 01 \rightarrow$ back to $00 \rightarrow$ repeat

- a) Draw the state diagram of this system. [2]
- b) Draw the state table with the input function for the T flip-flops. [2]
- c) Derive the minimized logic equations for the flip-flop inputs. [2]
- d) Draw the circuit diagram for this system. [2]

2. The Chittagong Port Authority (CPA) is installing an automated lock-control indicator for small-craft channels at the New Mooring Dock. The indicator cycles through 8 distinct display patterns to guide incoming vessels during docking. To save space and power, the engineering team asks an undergraduate group from UIU to design a synchronous digital counter using D flip-flops that implements the required pattern sequence. $0 \rightarrow 5 \rightarrow 7 \rightarrow 6 \rightarrow 4 \rightarrow 1 \rightarrow 2 \rightarrow 0$ (and then repeat)

Ensure any unused states are treated as don't-cares. To accommodate the design requirements, the coordinator of the project, Professor Wang, has instructed the students to complete the following tasks:

- a) The state table, the required sequence of display states (encoded as 3-bit binary numbers) [3]
- b) The optimized input functions, and [3]
- c) The circuit diagram. [2]

3. a) Asif is late for his DLD lab mid-term exam. When he reached the lab, he found that only 2-to-4 line decoders were available. However, for his task, he needed a 3-to-8 line decoder to complete his circuit. Explain how Asif can use two 2-to-4 decoders to implement a 3-to-8 decoder. [4]

- b) Implement the function $F(A, B, C, D) = \prod(1, 2, 3, 7, 8, 9, 11, 12, 14, 15)$ using a 4x1 MUX. [4]

4. Design a 3-bit universal shift register using D flip-flops. The register should perform different operations based on the three control signals S2, S1, and S0, as described in the function table below. [8]

If the control inputs do not match any of the listed combinations, the register contents must remain unchanged.

S2	S1	S0	Operation
0	0	1	Left Shift
0	1	1	Toggle
1	0	1	Set
1	1	0	Right Shift

- 5 At a busy intersection, multiple emergency and priority signals may arrive simultaneously. You want a system that decides which signal to attend to first based on predefined priorities, so the controller always responds to the most urgent vehicle.

Each type of vehicle is assigned a letter input and a 3-bit code (representing the signal in binary). The system outputs the highest-priority signal that should be handled first.

Priority	Vehicle	Input - Output
1 (highest)	Ambulance	A - 010
2	Fire Truck	F - 011
3	Police	P - 111
4	School Bus	S - 001
5 (lowest)	VIP Convoy	V - 110

You have to answer the followings:

- Derive the truth table of the priority encoder including the valid bit. [3]
- Derive the Boolean expressions for all the outputs. [3]
- Draw the logic diagram using basic gates. [2]

Excitation Tables for different Flip-Flops

Q(t)	Q(t+1)	D	Operation
0	0	0	Reset
0	1	1	Set
1	0	0	Reset
1	1	1	Set

Q(t)	Q(t+1)	T	Operation
0	0	0	No change
0	1	1	Complement
1	0	1	Complement
1	1	0	No change